MILITARY SPECIFICATION

REMOVER, AIRCRAFT CANOPY, RAU-1/A ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Remover, Aircraft Canopy, RAU-1/A, Assembly, loaded with Cartridge, Aircraft Canopy Remover, CCU-28/A for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MS-28741	- Hose Assembly - Detachable End Fitting, Medium Pressure

DRAWINGS

Armament Command

F11751080	- Remover, Aircraft Canopy, RAU-1/A, Assembly
C8797385	- Box, Fiberboard (Inner) Packing, Ammunition for Remover, Aircraft Canopy, M3A1 and RAU-1/A

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2.1 (Cont'd)

C8797386	- Box, Packing, Ammunition for Remover, Aircraft Canopy, M3A1 and RAU-1/A
C9235844	- Carton, Alternative Packing, Ammunition, for Remover, Aircraft Canopy, M3A1 and RAU-1/A
LI 11751080	- Index to Inspection Equipment List for Remover, Aircraft Canopy, RAU-1/A, Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-Material, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Torque.-Torque requirements shall be as specified in Table I. No loosening or deformation shall occur upon application of the breakaway torque.

TABLE I

Component	B Seating or Bottoming (Inch pounds)	C Breakaway (Inch Pounds)
Block and inside tube after insertion of cartridge	400 + 50	300 min.
Ring, stop and outside tube	500 + 50	400 min.

3.3 Assembly.

3.3.1 The body face surface of the plug shall be in complete contact with the cartridge.

3.3.2 The block shall be free to rotate 360 degrees.

3.4 Residual magnetism.-The assembled remover shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.5 Functional.

3.5.1 Ballistics.-Removers loaded with accepted CCU-28/A cartridges shall propel a 362 pound mass vertically and shall comply with the requirements listed in Table II.

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TABLE II

Temperature	Velocity (See 6.3) (min)	Thrust (min.)	Ignition Delay (See 6.2)(max)
-65°F to -70°F	25 fps	5400 lbs.	40 milliseconds
65°F to 75°F	25 fps	5600 lbs.	40 milliseconds
200°F to 205°F	25 fps	6100 lbs.	40 milliseconds

3.5.1.1 Initiation.-The remover shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 10 feet of aircraft hose, size no. 4, MS 28741 at 70° ± 5°F.

3.5.2 The remover shall not fail to fire.

3.5.3 The remover shall function without mechanical failure (See 6.1).

3.6 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.-The component shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defects.

3.6.2 Burr.-The component shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.-The component or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

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4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.-A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of accepted cartridges.

4.2.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table III. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table III. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE III

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3
1801 - 10000	200	2	3	3	4

4.2.2.1 Cartridge, Impulse, CCU-28/A, Drawing F11751078, covering a detail of Drawing F11751080.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. "O" ring missing	Visual
4. Sealing disc missing, damaged or improperly secured	Visual
5. Marking missing, incorrect or illegible	Visual
MAJOR	
101. "O" ring split, cut or cracked	Visual
102. Foreign matter	Visual

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4.2.2.1 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Case scratch, wrinkled, bulged or dented	Visual
202. Varnish missing or inadequate	Visual

4.2.2.2 Seal, Drawing C8593665, covering a detail of Drawing F11751080.

CRITICAL

None defined

MAJOR

None defined

MINOR	
201. Inside diameter under flange	Gage
202. Inside diameter adjacent to flange	Gage
203. Radius missing or incorrect	Visual
204. Metal defective	Visual
205. Protective finish inadequate or defective	Visual
206. Burr	Visual
207. Foreign matter	Visual

4.2.2.3 Assembly, Drawing F11751080, less all drawings except C8593644, B8593654, C8593657, A8593663, C8593709, D8593714, C8593715 and AN6227-11.

CRITICAL	
1. Gas port hole obstructed	Visual
2. Any part missing	Visual
3. Improper assembly	Visual
MAJOR	
101. Thread damaged	Visual
102. Burr	Visual
103. Foreign matter	Visual

MINOR

None defined

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4.2.2.4 Assembly, Drawing F11751080, less drawings B8593660, C8593665, D8593703, C8593707 and C8593711.

Defect	Method of Inspection
CRITICAL	
1. Body face surface of plug not in complete contact with cartridge	Gage
2. Cartridge missing	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.2.2.5 Assembly, Drawing F11751080, less all drawings except B8593660, D8593703, C8593707 and C8593711.

CRITICAL	
1. Seal on bearing tube missing	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.2.2.6 Assembly, Drawing F11751080, (with stop ring not engaged with outside tube).

CRITICAL	
1. Inside tube not free to slide	Manual
MAJOR	
None defined	

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4.2.2.6 (Cont'd)

Defect	Method of Inspection
MINOR	
None defined	
4.2.2.7 Assembly, Drawing F11751080.	
CRITICAL	
1. Mixed removers	Visual
2. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Diameter of block mounting hole	Gage
102. Diameter of cap mounting hole	Gage
103. Block not free to rotate 360° in both directions	Manual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Length between mounting holes	Gage
202. Total length, max.	Gage
203. Location of labels	Visual
204. Shipping plug missing	Visual
205. Seal missing or insecure	Visual
206. Protective finish inadequate or defective	Visual
4.2.2.8 Packing in Fiberboard box, Drawing C8797385.	
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual

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4.2.2.8 (Cont'd)

	Defect	Method of Inspection
204.	Cut or damaged through all impregnated layers	Visual
205.	Marking missing, incorrect or illegible	Visual
206.	Tape incomplete or badly wrinkled	Visual
207.	Length of tear tab, min.	Visual
208.	Color of tape incorrect	Visual

4.2.2.9 Box, Packing (Level A), Drawing C8797386.

CRITICAL

None defined

MAJOR

None defined

MINOR

201.	Contents loose	Manual
202.	Hardware missing, loose, broken or improperly engaged	Visual/Manual
203.	Strapping missing, loose, broken or improperly assembled	Visual/Manual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of units incorrect	Visual
208.	Car seal missing, unsealed or improperly positioned	Visual

4.2.2.10 Carton, Packing (Level B), Drawing C9235844.

CRITICAL

None defined

MAJOR

None defined

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4.2.2.10 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual
202. Contents of carton exposed	Manual
203. End loose or distorted	Visual/Manual
204. Cut or damaged through all layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete	Visual

4.2.3 Inspection testing.-The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.-The breakaway torque shall be applied to each threaded joint listed in 3.2. Failure of any remover to comply with the requirements of 3.2, Table I shall be cause for rejection of the remover.

4.2.3.2 Residual magnetism.-Each remover shall be tested for excessive residual magnetism. Failure of any remover to comply with the requirements of 3.4 shall be cause for rejection of the remover.

4.2.3.3 Functional.-The following tests shall be performed by the Government:

4.2.3.3.1 Ballistics.-Twenty-one units shall be tested as specified in 4.3.3.1 to determine compliance with 3.5.1, 3.5.2 and 3.5.3. The units shall be temperature conditioned as specified in Table IV and 4.3.3.1.1. Failure of any unit to comply with the requirements shall be cause for rejection of the lot.

TABLE IV

Temperature	Sample Size
-65° to -70°F	7
65° to 75°F	7
200° to 205°F	7

4.2.4 Inspection equipment.-The examination and tests shall be made using equipment listed on LI 11751080.

4.3 Test methods and procedures.

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4.3.1 Breakaway torque.-A torque wrench, set at the torque requirement specified in 3.2, shall be used to determine the adequacy of the joint. The assembly shall be secured in the fixture prior to application of the torque.

4.3.2 Residual magnetism.-The remover shall be tested for excessive residual magnetism, using an approved compass, in an area free of local magnetic effects, by placing the remover 6 inches from and at the same height as the compass, in a north-south horizontal position (block end at south) with the compass set in an east-west heading. The remover shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 180 degrees in the horizontal position about its longitudinal axis and moved slowly bact past the compass.

4.3.3 Functional.

4.3.3.1 Ballistics.-The remover shall be rigidly supported in the fixture by the mounting points and initiated by a source delivering an output equal to a M3A1 Initiator connected to the remover with 10 feet of aircraft hose, size 4, as specified in MS-28741. The remover shall comply with the requirements of 3.5. The remover shall be tested within 10 minutes after removal from the conditioning chamber. When this time limit is exceeded, the remover shall be reconditioned at the original temperature.

4.3.3.1.1 Temperature conditioning.-A simulated M3A1 Remover, loaded with a dummy CCU-28/A Cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Level A.-Packing and marking shall be in accordance with Drawings C8797385 and F8797386.

5.2 Packing and marking.- Level B.-Packing and marking shall be in accordance with Drawings C8797385 and C9235844.

5.3 Packing and marking shall be as specified by the procuring activity.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

6.2 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design failure.

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6.3 "Ignition delay" is defined as the time that elapses between the firing of the cartridge primer and the beginning of the continuous rise on the thrust time record.

6.4 Velocity applies at tube separation.

6.5 Loaded Assembly Hazard Information:
Explosive Hazard Classification: Class 4
Storage Compatibility: Group F
Department of Transportation (DOT) Hazard Class B
DOT Marking - Explosive Power Device - Class B
Handle Carefully - (Keep Fire Away)

6.6 Hazard notice.-The Remover Assembly described herein and certain of its components are flammable and/or explosive, consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

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Preparing Activity:

Army - MU

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